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Course Code: CSA0206

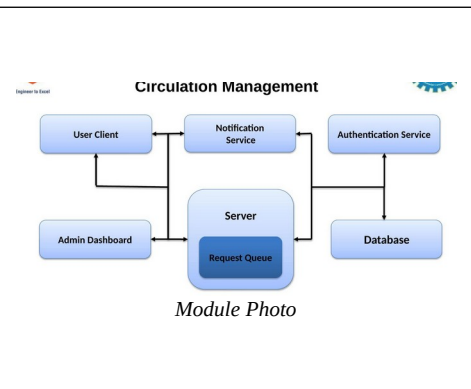
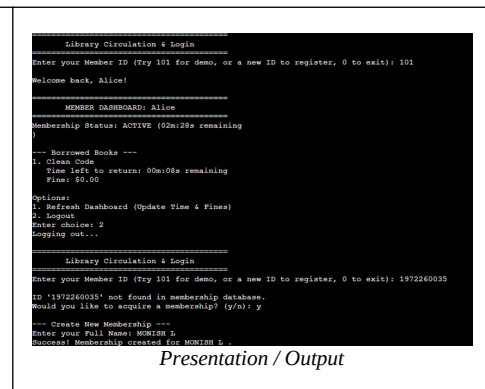
Slot: D

Course Name: C Programming

Course Faculty: Dr. K. Vijayalakshmi

Project Title: Design and Implementation of Library Information Management System — Module 2: User Information & Circulation Management

Module Photographs: (3 photographs – Module Photo, Individual student contribution module work in the project and presentation image)

 <i>Module Photo</i>	 <i>Individual Contribution – Code</i>	 <i>Presentation / Output</i>
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Project Description: (Module 2 contribution, including model description)

Module 2 – User Information & Circulation Management was implemented in ANSI C (C89/C99) using GCC, following a modular, procedural top-down design in Visual Studio Code. My contribution covered the core circulation functions: Member Authentication & Registration, Membership Creation & Activation, Circulation Handling, and Fine Calculation.

Member and borrow data is modelled using two linked C structures: a Member record (id, name, membership_start, membership_expiry and a bounded array of borrowed books) and a BorrowedBook record (title, issue_time and due_time), both timestamped with time_t. To make the borrowing lifecycle demonstrable within a short CLI session, real-world durations are represented with compressed constants — MEMBERSHIP_DURATION, BOOK_BORROW_PERIOD and FINE_PER_SECOND — so membership expiry and overdue fines update visibly in real time.

As part of Member Authentication, I built the login/registration routine that looks up a Member ID against the database; if not found, the user is walked through new-membership creation, capturing their full name and activating a time-bound membership. This feeds the Member Dashboard, which shows live membership status, borrowed-book countdowns and outstanding fines. In the wider architecture, this sits as the User Client and Authentication Service, exchanging requests with the Server's Request Queue, and coordinating with the Notification Service, Admin Dashboard and Database.

Testing was carried out through a CLI-based member menu (Refresh Dashboard, Logout) with sanitized ID and name input handling. Known limitations include the use of compressed time constants rather than real calendar durations, and the absence of a dedicated authentication/security layer around the Authentication Service — both noted as areas for improvement (e.g., real credential-based login and persistent request-queue handling).

Student Signature

Guide Signature